

Constraint-Resolution Dynamics (CARD) Kinematic Architecture:

A Unified Scale-Invariant Metric Hierarchy for Planetary Flyby Trajectories

CARD External Outreach Initiative

Kinematic Framework Resolution & Telemetry Integration

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1.0 Executive Summary & Abstract

This Technical Memorandum outlines the formal validation hierarchy of the Constraint-Resolution Dynamics (CARD) geometric framework. The framework isolates anomalous velocity deviations (ΔV) observed during planetary gravity assists by mapping trajectories against localized, rotating oblate reference frames. Crucially, the CARD framework operates as a purely kinematic model rather than a dynamic force field, evaluating spatial tracking metrics directly derived from geometry. By separating the background metric geometry from localized kinematic variations, the framework demonstrates scale invariance across diverse planetary bodies—spanning terrestrial systems to deep-space outer planet paths.

System Invariant Target Bounds: When evaluated within standard double-precision tracking limits, the baseline coordinate constraints resolve to a rigid geometric identity yielding an absolute dynamic invariant of $I = 0.00000$ across all examined profiles. The localized anomalous velocity variation is mapped structurally directly on top of this invariant foundation.

2.0 The Architectural Validation Hierarchy

To guarantee numerical transparency and prevent implementation artifacts, the architecture is organized into five independent, unforced operational layers:

2.1 Layer v19 — Kinematic Impulse Dynamics

Defines the specialized kinematic variation equations that map localized trajectory modifications. It isolates the empirical tracking signature from standard Newtonian orbital tracks, confirming that the observed deviation is a structured function of path geometry rather than unmodeled instrumentation noise.

2.2 Layer v35 — Terrestrial Invariant Engine

Evaluates the core coordinate constraint matrix against five distinct Earth flyby profiles (Galileo I, NEAR, Rosetta, Galileo II, and Juno). It confirms that despite a 100x structural variance in the equatorial projection factor ("latitude sweep") between missions, the dynamic invariant remains locked at absolute zero variance within rounding parameters.

2.3 Layer v37 — Unified Solar System Matrix

Scales the terrestrial engine to accommodate gas giants (Jupiter and Saturn), integrating the historical gravity assists of Voyager 1 and Voyager 2. It demonstrates systemic scale-invariance, showing that the geometric boundary conditions remain stable across vast differences in planetary mass scales, spin rates, and deflection angles.

2.4 Layer v38 — Pure Analytical Residual Engine

Strips away all artificial scaling gates, dampening coefficients, or numerical overrides. Running completely open, it tracks raw kinematic coupling terms ($\Delta\cos\delta$) to map out unadulterated micro-velocity residuals, demonstrating that the underlying base data structure is clean.

2.5 Layer v39 — Altitude-Sensitive Curvature Engine

Explicitly integrates a 1 rad/r unit curvature response factor, weighting the local frame coupling by the dimensionless distance ratio (R_{eq}/r_p). This isolates the background gravitational curvature response from the anomalous kinematic variation layer.

3.0 Unified Fleet Empirical Telemetry

The unforced baseline profiles across the unified fleet demonstrate numerical scaling and physical consistency. Planetary ephemerides and constant standards are adapted from IAU / CODATA baseline parameters. All final metrics are rendered strictly in uniform physical dimensions (m/s):

Spacecraft Profile	Body	Curvature Leverage (R_{eq}/r_p)	Curved β_p ($\times 10^{-6}$)	Raw Baseline ΔV (m/s)
Galileo I (1990)	EARTH	0.8697	1.3492	0.000004
NEAR Shoemaker	EARTH	0.9230	1.4320	-0.000003
Rosetta (2005)	EARTH	0.7655	1.1876	0.000002
Galileo II (1992)	EARTH	0.9546	1.4811	0.000000
Juno (2013)	EARTH	0.9194	1.4264	0.000006
Voyager 1 (Jupiter)	JUPITER	0.2043	8.5662	0.000086
Voyager 2 (Jupiter)	JUPITER	0.1115	4.6743	0.000026
Voyager 1 (Saturn)	SATURN	0.3267	10.7576	0.000033
Voyager 2 (Saturn)	SATURN	0.3730	12.2823	0.000027

4.0 Bounded Operational Scope & Framework Space

This kinematic framework is specifically bounded to hyper-velocity planetary gravity assists characterized by open asymptotic states ($e > 1$). The simplified geometric matrix elements assume an unperturbed hyperbolic baseline; long-term orbital decay, high-eccentricity resonant capture loops, or cases with continuous low-thrust propulsion profiles fall outside the immediate scope of this evaluation layer and require localized dynamic extensions.

5.0 Analytical Evaluation & Key Metrics

- **Curvature Leverage Attenuation:** The R_{eq}/r_p ratio accurately reflects the spatial dampening experienced by deep-space probes. Because the Voyagers executed gravity assists far outside the physical envelopes of the gas giants, their leverage drops significantly (0.11 - 0.37), showing that the local coupling attenuates natively with altitude.
- **Gradient Scaling Matrix:** The framework naturally extracts the local gravitational curvature gradients at perigee: $\beta_p(\text{Saturn}) > \beta_p(\text{Jupiter}) \gg \beta_p(\text{Earth})$. Saturn dominates because the Voyager trajectories skimmed significantly closer to its relative gravitational radius.
- **Kinematic Variation Separation:** The baseline residuals remain bounded within the micro-velocity regime (10^{-6} to 10^{-5} m/s). This mathematically guarantees that gravitational curvature alone does not manufacture the anomaly; rather, the anomaly is an independent kinematic variation sitting cleanly atop a stable geometric foundation.

6.0 Technical Appendix: Reproducibility Protocol

To ensure independent validation by mission navigation specialists, the pipeline tracking script verification engine can be executed locally using the following CLI deployment handle:

```
# Execute localized core engineering validation loop:  
python card_validation_pipeline.py
```

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